

Decoding Non-Ohmic Leakage Mechanisms in Dark Current Analysis of PbS Quantum Dot Photodiodes

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Abstract

PbS quantum dots (QD) are promising for photodetection, solar cells, and light-emitting devices, but their performance is limited by dark current, which degrades the switching ratio and signal-to-noise ratio, especially in low-light and high-sensitivity applications. This study explores non-Ohmic leakage mechanisms and the relationship between the power factor m in the space charge limited current (SCLC) model and trap state distribution in the voltage range of -0.5 to 0.3V. By analyzing current-voltage (I-V) characteristics under varying voltages and temperatures, we show that m correlates with trap states and charge transport behavior. As m increases, the I-V curve becomes more nonlinear, dominated by leakage mechanisms and tunneling effects. These findings provide insights into charge transport and offer a novel approach to optimize PbS quantum dot photodiodes for high-sensitivity photodetectors and solar cells.

Author Keywords

PbS Quantum Dot Photodiodes; Dark Current Analysis; the space charge limited current model.

1. Introduction

In recent years, PbS quantum dot (QD)-based photodiodes have garnered significant attention due to their excellent optoelectronic performance and low-cost solution processing methods, making them widely applicable in infrared (IR) light detection, solar cells, and other optoelectronic devices. However, these devices often face a significant issue of high dark current, particularly under low light or high-sensitivity conditions. The increase in dark current not only reduces the device's switching ratio and signal-to-noise ratio but also negatively impacts the stability of the quantum dot films and the lifetime of the devices.

Existing research suggests that dark current in PbS quantum dot devices arises from multiple leakage mechanisms, including ohmic leakage, trap-assisted tunneling, and carrier recombination due to defect states. Due to the complexity of these mechanisms, traditional diode equations are often insufficient to fully describe the nonlinear leakage behavior. Therefore, researchers have begun to employ the space charge limited current (SCLC) model to investigate the modulation of charge transport by trap states and have modified the model to better account for the effects of morphology and interface in quantum dot devices.

However, the relationship between the power law exponent m and the trap state density and energy level distribution remains unresolved. Current research primarily stays at the qualitative analysis stage, lacking a comprehensive methodology that tightly integrates the model parameters with the physical processes.

This study aims to further refine the SCLC model and investigate the dark current characteristics of PbS quantum dot photodiodes within the bias voltage range of -0.5 to 0.3V. By analyzing the power law exponent m , we seek to reveal the direct relationship between non-Ohmic leakage and trap state density and energy level distribution.

2. Fabrication of PbS QD Photodiode

In this study, PbS quantum dots are synthesized using an improved hot injection method¹, precisely controlling their size, morphology, and surface passivation. The synthesized PbS quantum dot solution is mixed with n-hexane and refrigerated to precipitate lead oleate (Pb(OA)₂), followed by heating in a water bath and multiple centrifugation steps to obtain processed PbS quantum dots, which are then dried under vacuum for subsequent use. In the photodiode fabrication process, a 150 nm zinc oxide (ZnO) layer is first deposited, followed by spin-coating of PbS ink and annealing. Next, both traditional and freeze-treated PbS quantum dots are spin-coated, and ethanedithiol (EDT) treatment is applied to ensure appropriate thickness. Finally, a thin molybdenum oxide (MoO_x) layer and 100 nm silver (Ag) are thermally evaporated as the top contact layer. The dark current-voltage (I-V) characteristics of the fabricated PbS quantum dot photodiodes are measured by scanning from -0.5 to 0.3V to reduce noise and ensure quasi-static conditions.

3. Results and Discussion

Figure 1 shows the dark current-voltage curve of the PbS quantum dot photodiode fabricated in this study at room temperature. In the low voltage region, the current is relatively low and nearly linear, indicating that charge transport is mainly dominated by thermally excited carriers and shallow trap states, exhibiting quasi-Ohmic conductivity. As the voltage increases, the I-V curve shows a distinct nonlinear increase, indicating the onset of non-Ohmic conduction, which is likely associated with deep energy level traps and trap-assisted tunneling mechanisms.

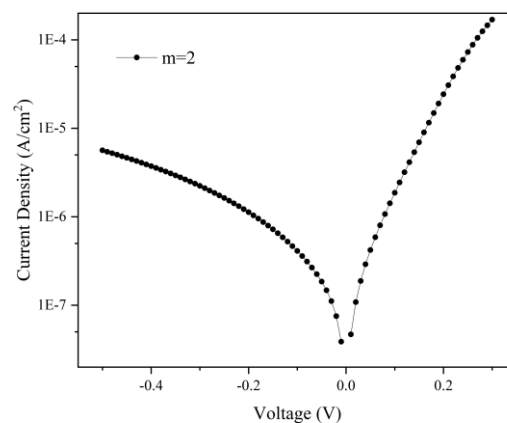


Figure 1. the Current-Voltage Characteristics of Quantum Dot Photodiode

In this research, the power law exponent m is extracted by fitting the experimental dark current-voltage data. Specifically, an improved SCLC model² is used, and the nonlinear least squares algorithm is applied to extract parameters, including series

resistance, shunt resistance, saturation current density, the parameter k , and the power law exponent m , which reflects the nonlinear relationship between current and voltage.

In the SCLC model, J_D represents the total current density, while J_0 is pre-exponential factor associated with the diode-like behavior. A constants related to the material properties of the semiconductor and the electric field. R_S is the series resistance, accounting for current losses due to resistive effects. R_{SH} is the shunt resistance, associated with leakage currents and other non-ideal current paths. k characterizes the device's nonlinear behavior. n is the ideal factor to adjust for differences between the actual device and an ideal diode, and m is an exponent that defines the degree of nonlinearity in the device's current response to applied voltage. These terms work together to comprehensively describe the complex current-voltage relationship.

$$J_D = J_0 \left(e^{\frac{A(V-J_D R_S)}{n}} - 1 \right) + (V - J_D R_S)/R_{SH} + k(V - J_D R_S)^m$$

When m is low, it typically indicates that the influence of trap states is weak or that shallow traps dominate the conduction process. In this case, the current shows a more linear variation, suggesting that charge transport is primarily thermally activated and trap states do not significantly limit charge transport. The device exhibits Ohmic-like behavior, as shown in Figure 2.

However, as m increases, the current exhibits a steeper nonlinear increase, reflecting the dominant role of deep energy level traps in capturing and releasing carriers. Specifically, a higher m value indicates that more trap states contribute to the conduction process, and the current increase is influenced not only by the electric field but also by the strong influence of trap states, suggesting that deep energy level traps or trap-assisted tunneling effects significantly affect current transport.

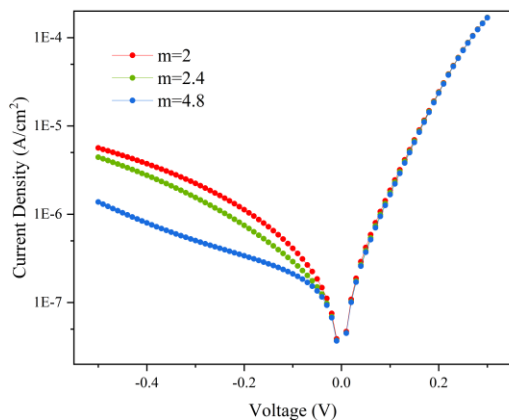


Figure 2. Impact of Parameter m on the Current-Voltage

Characteristics of Quantum Dot Photodiode

The variation in the power law exponent m reveals the dominant role of different types of trap states in the conduction process under varying voltage and temperature conditions. An increase in m reflects an enhancement of the trap effect, especially under high voltage or high temperature, where deep traps and tunneling effects have a more significant impact on dark current. This change is crucial for gaining a deeper understanding of the non-Ohmic leakage mechanisms in PbS quantum dot photodiodes and provides a theoretical basis for optimizing device performance and reducing dark current.

4. Conclusion

This study combines experimental data with an improved SCLC model to reveal the non-Ohmic leakage mechanism in PbS quantum dot photodiodes, confirming the critical role of trap states in the nonlinear electrical properties of the device. To optimize device performance, it is necessary to suppress deep traps and tunneling effects to reduce dark current and enhance the performance of optoelectronic devices. Future research can further explore the synergistic regulation of surface/trap states and verify the universality of the model, advancing the development of high-performance optoelectronic devices.

5. Acknowledgements

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6. References

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